

Vehicle Parking System - Architecture Selection & Justification

Architecture Selection

We chose **Layered Architecture** for the **Vehicle Parking System**.

Architectural Choice

The system is divided into separate layers: **Presentation, Business, and Data**. The Presentation Layer handles user and administrator interactions, the Business Layer handles parking, vehicle, reservation, payment, and fee processing, and the Data Layer stores user, vehicle, parking, and payment information.

Reason 1

Layered Architecture provides **clear separation of concerns**, making the different functions of the Vehicle Parking System easier to organize and maintain.

Reason 2

The system has distinct responsibilities such as **user management, vehicle registration, parking slot management, vehicle entry and exit, fee calculation, payment processing, and data storage**, which can be organized into appropriate layers.

Security Advantage

Separating the **presentation, business, and data layers** prevents the user interface from directly accessing stored data, providing better control over data access and helping protect user and parking information.

Performance Benefit

Separating responsibilities between layers allows the system to process **parking requests, slot availability checks, fee calculations, and data operations in an organized manner**, avoiding unnecessary interaction between unrelated components.